CALCULUS · UNIT 8: DIFFERENTIAL EQUATIONS

Cold Case: Newton's Cooling

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students solve a forensic mystery with Newton's Law of Cooling. They read slope fields, approximate with Euler's method, solve the separable equation, and use two temperature readings to estimate when the cocoa was poured.

Measurement uncertainty tests whether the conclusion is robust.

Standards Addressed

- AP FUN-7 Differential equations: verifying solutions, slope fields
- AP FUN-7.D Euler's method (BC topic as enrichment)
- AP FUN-7.E Separation of variables and initial conditions
- AP FUN-7.F–H Exponential (and logistic) models

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the mystery. Part 1 rates and slope fields.
Day 2	Part 2 Euler and separation of variables.
Day 3	Part 3 case analysis and the verdict.

Materials

- Student handout and Evidence Cards
- Graphing calculators
- Optional: a mug of hot water and thermometer to collect real data

How the Three Versions Differ

Level	What changes
CP2	Signs of rates, verifying a solution, slope field, and solving for k and time with a given model.
CP1	Euler's method, separation of variables, second derivative, and uncertainty.
Honors	Stability, Euler error, non-separable extension, logistic model, and testing an alibi.

Watch For (Common Misconceptions)

- Losing the absolute value or sign when integrating $\frac{1}{T-68}$.
- Using minutes and hours inconsistently.
- Forgetting that t is negative before the first reading.

Extension Ideas

- Collect cooling data for hot water and fit k.
- Research how forensic scientists estimate time of death.
- Model warming of a cold drink.

Answer Key – CP2

Part 1: Rates of Cooling

#	Task	Answer
1	Is $\frac{dT}{dt}$ positive or negative when the cocoa is 140°F? What does that mean?	Negative; the cocoa is cooling.
2	Where is the cocoa cooling fastest: at 140°F or 100°F? Why?	At 140°F; the difference from the room is larger.
3	Verify that T = 68 + 72e^(−kt) satisfies $\frac{dT}{dt}$ = −k(T − 68).	$\frac{dT}{dt}$ = −72ke^(−kt) = −k(T − 68) ✓
4	What happens to T as t becomes very large?	T approaches 68°F (room temperature).

Part 2: Slope Field

#	Task	Answer
5	Find the slope at T = 140, T = 104, T = 68, and T = 50.	−1.44, −0.72, 0, 0.36
6	Sketch the slope field for 0 ≤ t ≤ 60 and 40 ≤ T ≤ 160. Describe the pattern.	Slopes depend only on T; horizontal at 68; steeper farther from 68.
7	Sketch the solution through (0, 140).	A decreasing curve that levels toward 68
8	Explain why the line T = 68 is a solution.	If T = 68, the rate is 0, so the temperature does not change.

Part 3: Solve the Case

#	Task	Answer
9	Find k.	72e^(−30k) = 48 → k = $\frac{\ln 1.5}{30}$ ≈ 0.0135 per minute
10	When was the cocoa 185°F? Solve 68 + 72e^(−kt) = 185 (t will be negative).	e^(−kt) = $\frac{117}{72}$ → t = −$\frac{\ln 1.625}{0.0135}$ ≈ −36 min → about 8:24 p.m.
11	Which suspect was in the room when the cocoa was poured?	Suspect A (8:15–8:35 p.m.)

Reflection

#	Prompt	Look for
R1	Why does the cocoa cool more slowly over time?	The rate is proportional to the difference from room temperature, which gets smaller.

#	Prompt	Look for
R2	Why was t negative when solving for 185°F?	The cocoa was poured before the first reading at 9:00 p.m.

Answer Key – CP1

Part 1: Euler's Method

#	Task	Answer
1	Use Euler's method with Δt = 10 min to estimate T(30).	T(10) ≈ 140 − 9.72 = 130.28; T(20) ≈ 130.28 − 8.41 = 121.87; T(30) ≈ 121.87 − 7.27 ≈ 114.6
2	Compare with the measured 116°F. Is Euler's estimate an over- or underestimate? Why?	Underestimate; the solution is concave up, so tangent lines lie below the curve.
3	Sketch the slope field and your Euler path.	Answers vary; segments follow the field.
4	Find $\frac{d^2T}{dt^2}$ in terms of T and use it to explain the concavity.	$\frac{d^2T}{dt^2}$ = 0.0135²(T − 68) > 0 when T > 68 → concave up

Part 2: Separation of Variables

#	Task	Answer
5	Separate and integrate to find the general solution.	$\int\frac{dT}{T-68}$ = −∫k dt → ln|T − 68| = −kt + C → T = 68 + Ae^(−kt)
6	Use T(0) = 140 and T(30) = 116 to find A and k.	A = 72; k = $\frac{\ln 1.5}{30}$ ≈ 0.01352
7	Find when the cocoa was 185°F.	t ≈ −35.9 min → about 8:24 p.m.
8	Find when the cocoa reaches 80°F.	72e^(−kt) = 12 → t = $\frac{\ln 6}{k}$ ≈ 132.5 min → about 11:13 p.m.

Part 3: Case Analysis

#	Task	Answer
9	The thermometer may be off by ±2°F at 9:30 (114–118°F). Find the range of possible pour times.	116°F → 8:24. 114°F: k = $\frac{\ln(72\div46)}{30}$ ≈ 0.01493 → t ≈ −32.5 min → about 8:27. 118°F: k = $\frac{\ln(72\div50)}{30}$ ≈ 0.01215 → t ≈ −39.9 min → about 8:20. Range ≈ 8:20–8:27 p.m.
10	Does the uncertainty change which suspect is implicated?	No; the whole range is during Suspect A's visit (8:15–8:35).
11	Write the case file with the model, calculations, and conclusion.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why does separation of variables give an exponential solution?	Integrating $\frac{1}{T-68}$ gives a logarithm, and solving for T gives an exponential.
R2	How can measurement error affect a conclusion?	It changes the model's parameters, so you should check whether the conclusion holds for the whole range.

Answer Key – Honors

Part 1: Qualitative Analysis

#	Task	Answer
1	Classify the equilibrium T = R as stable or unstable. Explain with the slope field.	Stable; solutions above decrease toward R and below increase toward R.
2	Show that every solution approaches R and find $\frac{dT}{dt}$ when T = R + 20 in terms of k.	T = R + Ae^(−kt) → R; $\frac{dT}{dt}$ = −20k
3	Use Euler's method with Δt = 15 min from T(0) = 140 to T(30) (k ≈ 0.0135). Find the error compared with 116.	T(15) ≈ 125.40; T(30) ≈ 113.77; error ≈ 2.2°F
4	Explain how halving Δt changes the error.	For Euler's method the error is roughly proportional to Δt, so it roughly halves.

Part 2: Analytic Solution

#	Task	Answer
5	Solve the IVP and find the pour time.	T = 68 + 72e^(−0.01352t); pour at t ≈ −35.9 min → about 8:24 p.m.
6	The room heater turned off at 9:00 p.m. and the room cooled: R(t) = 68 − 0.05t. Write the new differential equation. Why is it no longer separable?	$\frac{dT}{dt}$ = −k(T − 68 + 0.05t); the right side cannot be written as g(t)h(T).
7	Rumor spreads in the museum by the logistic model $\frac{dP}{dt}$ = 0.4P(1 − $\frac{P}{200}$), P(0) = 10. Find P(t).	P = $\frac{200}{1+19e^{(-0.4t)}}$
8	When do half of the 200 staff know the rumor?	19e^(−0.4t) = 1 → t = $\frac{\ln 19}{0.4}$ ≈ 7.4 hours

Part 3: Verdict

#	Task	Answer
9	Using the thermometer uncertainty (±2°F at 9:30), find the range of pour times.	About 8:20 to 8:27 p.m.
10	Suspect A left at 8:35. Suspect B claims the cocoa was poured at 8:40. What temperature would the cocoa have had at 9:00 if poured at 8:40? Does this agree with the evidence?	185 → after 20 min: 68 + 117e^(−0.2704) ≈ 157.3°F ≠ 140°F; no
11	Write the case file with the model, assumptions, uncertainty, and verdict.	Answers vary.

Reflection

#	Prompt	Look for
R1	What assumptions does Newton's Law of Cooling make?	Constant room temperature and cooling rate proportional to the difference; the mug and liquid cool uniformly.
R2	How does a logistic model differ from exponential growth?	Growth slows as P approaches the carrying capacity.